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Empowering Primary Batteries: Tailoring Ionic Conductivity in Methylcellulose Based Solid Polymer Electrolytes via Aluminium Nitrate Doping

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This study presents the preparation of a solid polymer electrolyte film using a biopolymer, Methylcellulose, doped with aluminium nitrate ($Al(NO_3)_3.9H_2O$) via the solution casting method. The structural and morphological properties of the polymer electrolyte were examined using FTIR analysis, XRD, and Scanning Electron Microscopy. FTIR analysis provides the complexation between the polymer and the dopant. The XRD analysis accounts for the polymer's structural modification with the salt's inclusion. Impedance analysis indicates the enhancement of the electrical properties of the polymer. An electrolyte system with 30 wt% of the salt exhibits a room temperature ionic conductivity of 3.78×10^{-5} S cm^{-1}. I-V and I-t characteristics of the highest conducting electrolyte systems were carried out. In this work, a primary battery was designed with a state-of-the-art, high-conductivity electrolyte. The resulting battery's discharge characteristics were then evaluated.

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In this developing world, polymers have permeated every industry, from the packaging of food and beverages to the transportation sector to the medical sector. Polymers have a wide range of qualities that make them useful in several different ways, which determine their worth. Thus, polymers have become ubiquitous in our daily lives and are extraordinary materials that have revolutionized countless industries. Like monomers became the building blocks of polymers, polymers are the building blocks of modern society.[1–3] The development of polymer electrolytes has proven important in the twenty-first century. Polymer electrolytes are a remarkable class of materials at the forefront of technological advancement. Polymer electrolytes have a lot of applications, including batteries, fuel cells, and electric double-layer capacitors. It has been noted that during the past few decades, numerous in-depth studies have been conducted based on the synthesis and use of synthetic polymer-based polymer electrolytes, such as polyvinylidene fluoride (PVDF) and methylcellulose (MC), etc In addition to their wide range of uses, they have an advantageous quality of biodegradability. Most polymer research and development nowadays use biodegradable qualities while keeping an eye on a clean, green environment. There are primarily two categories of electrolytes: liquid and solid. Solid electrolytes dominate liquid electrolytes as they are less poisonous, non-flammable, and have superior safety and thermal stability than liquid electrolytes. They also reduce cell weight. As the name suggests, these solid-state electrolytes combine traditional liquid electrolytes' conductivity with the mechanical integrity of solid polymers.

Methylcellulose is one of the cellulose derivatives with 27.5%–31.5% methoxy groups (Fig. 1). The property of the polymer to form thermal gelation has increased its applicability.[4] It has a good film-forming ability with excellent thermal and mechanical properties.[5,6] The biodegradable polymer is widely used in coating, food, and drug industries.[7] Literature accounts for numerous works based on methylcellulose. Kadir et al.[8] prepared proton-conducting methylcellulose-based solid polymer electrolytes and obtained ionic conductivity of the order of 10^{-4} S cm^{-1}. Aziz et al.[9] synthesized methylcellulose-based polymer electrolyte systems and observed ionic conductivity of the order of 10^{-3} S cm^{-1}. The films exhibited excellent electrical properties. Jayalakshmi et al.[10] have prepared magnesium ion conducting solid polymer electrolyte systems, which showed good electrical, structural, and thermal properties.

Aluminium is among the most abundant elements in the Earth's crust, offering an impressive theoretical specific capacity of approximately 2.98 Ah g^{-1}. Its low electrochemical reactivity and stable passive oxide layer make it a particularly attractive candidate for energy storage systems, including primary (non-rechargeable) batteries. One of the key advantages of aluminium is its high volumetric capacity, which surpasses that of many other metal-based anodes, enabling compact battery designs with enhanced energy density. Additionally, the small ionic radius of the Al^{3+} ion (0.56 Å) facilitates efficient ion transport and intercalation processes, potentially improving electrode kinetics.[11]

From the perspective of primary battery fabrication, aluminium's cost-effectiveness, widespread availability, and lightweight nature contribute significantly to the economic and practical feasibility of large-scale production. Its environmental benignity compared to heavy metals like Mercury or cadmium also supports the development of more sustainable and safer disposable batteries. Moreover, the possibility of using aluminium as both an anode and a structural component can simplify cell architecture and reduce overall material usage.[12,13]

Kotobuki et al. prepared aluminium ion conducting solid polymer electrolytes based on PVDF doped with aluminium chloride salt. They obtained a maximum room temperature ionic conductivity of the order of 10^{-4} S cm^{-1}.[14] Leung et al. analyzed the electrochemical properties of solid polymer electrolytes based on 1-ethyl-3-methylimidazolium chloride and aluminium chloride. The prepared electrolyte system exhibited good electrochemical properties with an ionic conductivity of 13 mS cm^{-1}.[15] Mohammed et al. synthesized aluminium chloride incorporated Polyacrylonitrile (PAN) based electrolyte systems. They obtained an ionic conductivity of the order of 1.1 mS cm^{-1}.[16] Although various electrolyte systems have been extensively studied, there remains a noticeable lack of investigation into aluminum ion-doped electrolytes utilizing biopolymer frameworks, highlighting an opportunity for further research in this area.

Taking the excellent properties and broad literature of methylcellulose into account, we have made an attempt to prepare solid polymer electrolyte films based on methyl cellulose and aluminium salt. This study aims to explore the fascinating world of biopolymer electrolytes, dwelling on their synthesis, properties, and diverse applications while highlighting their pivotal role in shaping the future of electrochemical technologies.

Materials and Methodology

Materials.—Methylcellulose with low viscosity (Viscosity: 350–550 cps, 2% in water) and Aluminium nitrate nonahydrate 98% extra pure (Molecular weight: 375.13 g mol^{-1}) were procured from

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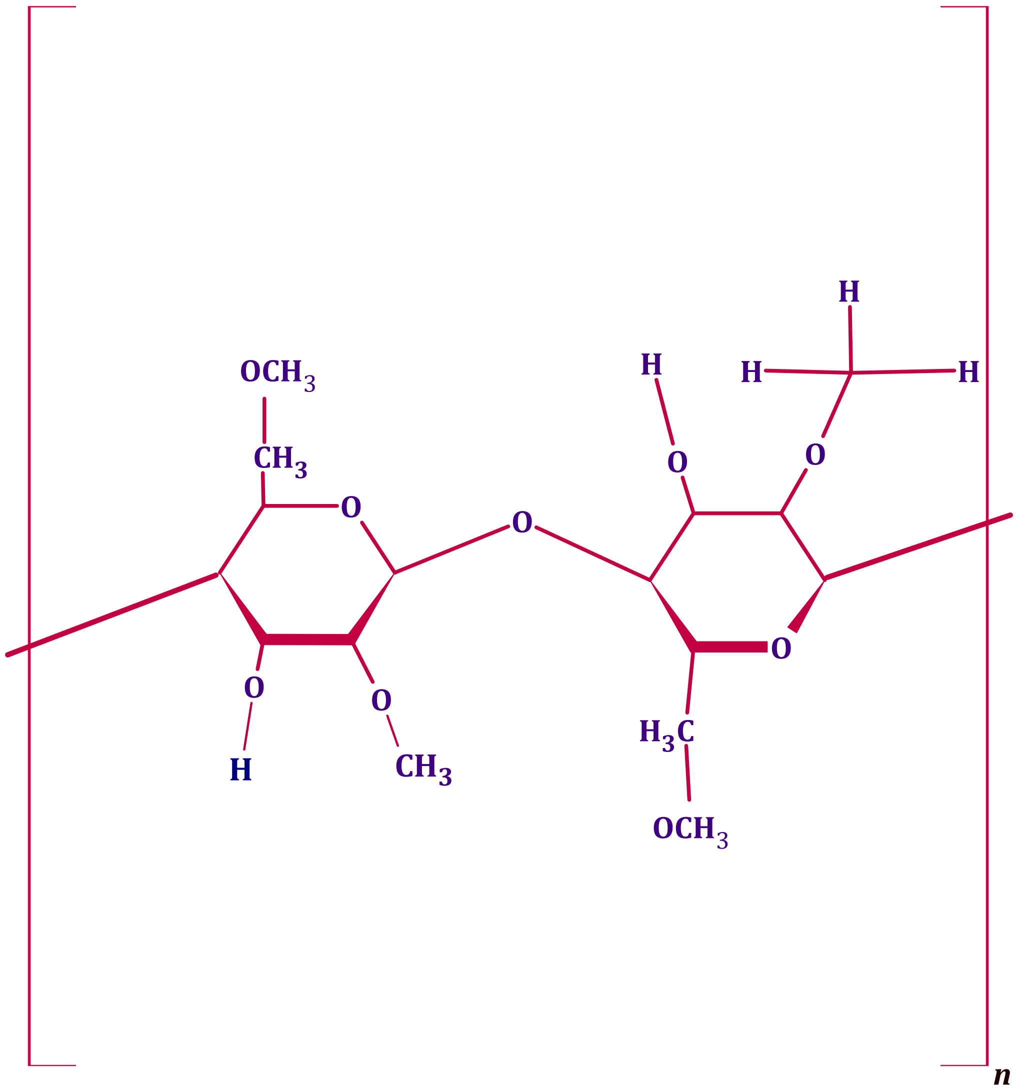

Figure 1. Structure of methylcellulose.

Loba Chemie Pvt. Ltd. and used as such. For fabrication of battery, Manganese dioxide (70%) was procured from Loba Chemie Pvt. Ltd., and graphite fine powder was purchased from Molychem.

Methodology.—This study utilized a solution-based processing technique to fabricate solid polymer electrolytes. Methylcellulose and Aluminium nitrate salt (total 2 g) were dissolved in deionized water (100 ml) at various weight percentages. Mixture of subjected to magnetic stirring at 40 °C for approximately 10 h to achieve a uniform solution. The solution was then cast into Petri dishes for slow solvent evaporation for 48 h, ultimately yielding solvent-free films. Following drying, the films were carefully detached and subjected to further drying in a hot air oven for 24 h at 60 °C to remove any residual moisture. Prior to additional characterization, the films were stored in a desiccator containing silica gel as a desiccant. The final polymer electrolyte sheets exhibited thicknesses ranging from 100 to 200 μm.[3,8] Table I illustrates the designations of the films with different compositions of aluminum salt. The electrolyte film with more than 30 wt% of the aluminum salt could not be prepared despite several attempts due to the high concentration of the salt.

Characterization techniques.—The structural analysis of different polymer electrolytes was carried out using Rigaku mini flex 5th generation XRD spectrometer using Cu-K$_{\alpha}$ radiation in the range of $2\theta = 5° - 80°$ with a step size of $2°\ \text{min}^{-1}$. The morphology of aluminium-doped methyl cellulose biopolymer electrolyte was found with the help of a Carl Zeiss scanning electron microscope. The prepared samples were subjected to the FTIR technique using a Shimadzu IR Spirit ATR-FTIR spectrometer. FTIR spectra were recorded between 400 cm^{-1} and 4000 cm^{-1} with the least count of 4 cm^{-1}. Impedance analysis was done with the help of an nF LCR in the frequency range of 40 Hz-5 MHz. Electrolyte films were sandwiched between two stainless electrodes of diameter 12 mm, and the measurement was taken in the frequency sweep mode by applying a 20 mV voltage. I-V and I-t characteristics were carried out with the help of Keithley source meter 2636B.

Preparation of primary battery.—An electrochemical cell (primary battery) was constructed utilizing a developed electrolyte formulation with a high ionic conductivity membrane. This setup aimed to assess the potential application of these electrolytes in electrochemical energy storage devices. The cathode was made up of a 3:1:1 mixture of graphite, vanadium pentoxide, and the highest conducting electrolyte film. This mixture was then thoroughly ground to produce a fine powder. Next, a pressure of about 5 tons was used to create the pellet. Aluminium metal was used as the anode. The electrolyte sheet was sandwiched between the anode and cathode and inside a custom-made battery holder. Subsequently, the prepared primary battery's open circuit voltage and discharge properties were examined.[3]

Table I. Designation of the prepared polymer electrolyte films.

Methyl cellulose (wt%)	Aluminium salt (wt%)	Designation
100	0	MA0
95	5	MA5
90	10	MA10
85	15	MA15
80	20	MA20
75	25	MA25
70	30	MA30

Results and Discussion

XRD analysis.—X-ray diffraction (XRD) was employed to investigate the crystalline structure of the aluminium-doped methylcellulose biopolymer electrolyte sheet. The XRD patterns (Fig. 2) revealed the complex nature of the dopant-polymer interactions. Two prominent peaks were observed in the spectra at around 9° and 20°. The peak at 9° suggests cellulose modification, while the broader peak around 20° indicates the amorphous character of methylcellulose.[17,18] As the concentration of salt increases, the polymer's crystallinity decreases, as evidenced by the peak at 20° being broader and its intensity declining.[3] The amorphousness of samples while increasing salt concentration increases due to the breaking and formation of hydrogen bonds.[9,19] The observed increase in the amorphous nature of the polymer electrolyte signifies potential complexation phenomena between the polymer chains and the introduced salt. This hypothesis is supported by X-ray diffraction (XRD) analysis, which can be used to quantify the extent of this change in crystallinity. Fityk software was then utilized to deconvolute the XRD patterns and determine the percentage of crystalline phase remaining in the polymer electrolyte.[20] Figure 3 depicts the results of crystallinity analysis using a deconvolution technique (Eq. 1).[8] The figure shows the deconvoluted graphs of the polymer electrolytes.[3]

$$\chi_c = \frac{A_c}{A_c + A_a} \times 100\% \qquad [1]$$

Where A_c and A_a are the areas under crystalline and amorphous curves. Table II lists the values of degree of crystallinity of the prepared electrolyte systems. Pristine methylcellulose has a crystallinity of 28.8%. The crystallinity of the methylcellulose electrolyte exhibits an inverse relationship with dopant concentration. This observation is consistent with the disruption of crystalline domains within the polymer matrix, potentially leading to enhanced ionic conductivity.[21] Among the samples analyzed, polymer electrolyte film MA30 exhibits the lowest crystallinity, at approximately 14.3%. This characteristic coincides with its demonstrably high ionic conductivity.

Fourier transform infrared spectroscopy (FTIR) analysis.—Fourier-transform infrared (FTIR) spectroscopy was employed to investigate the interaction between the functional groups within the polymer and the introduced salt. Figure 4 presents the FTIR spectra of various polymer electrolytes.

The spectrum of MA0 displays seven distinct peaks, each associated with a specific vibrational mode of its functional groups. The peak at 3447 cm^{-1} corresponds to the O-H stretching mode, indicating the presence of hydroxyl groups. The C-H asymmetric stretching mode is observed around 2905 cm^{-1}, while the O-H bending mode appears at 1649 cm^{-1}. The δ(C–H) deformation in the plane is signified by the peak around 1457 cm^{-1}. Additionally, peaks at 1339 cm^{-1} and 1052 cm^{-1} are attributed to C-H bending and C-O-C stretching modes, respectively. Finally, the sharp peak at 948 cm^{-1} suggests the presence of a glycosidic linkage.[3] Table III summarizes the peak positions associated with various functional group vibrations in all electrolyte systems. These peaks correspond to O-H asymmetric stretching, C-H asymmetric stretching, and C-H bending vibrations. An observable shift in these peaks is evident in the table. Generally, a shift in the FTIR peak of its disappearance indicates the complexation of the corresponding functional group with the constituents of the salt. Absence of δ(C-H) bond in the FTIR spectrum of MA10 and higher salt-containing systems indicates the complete complexation of the functional group of the polymer with the salt. The shift in the O-H asymmetric stretching bond also accounts for the complexation of the salt and polymer matrix.

The observed shift in peak position implies a potential interplay between the hydroxyl (O-H) and methyl (C-H) groups within the methylcellulose polymer and the constituent ions of the aluminium nitrate salt. This interaction can be attributed to the strong electron-withdrawing nature (high electronegativity) of the O-H group, facilitating its coordination with the positively charged aluminium (Al^{3+}) cations. Conversely, with a lower electronegativity, the C-H

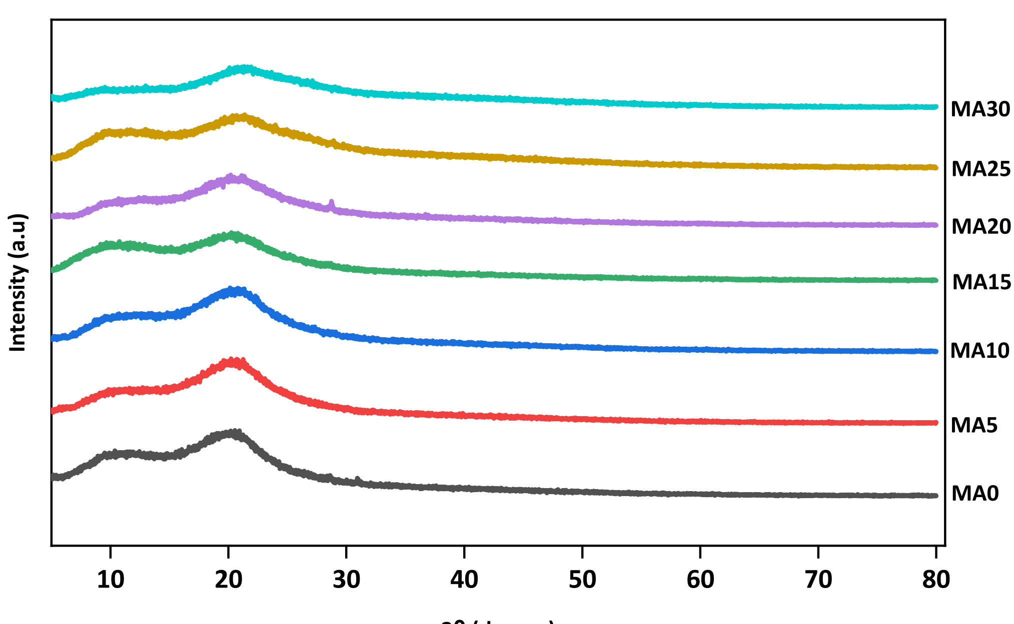

Figure 2. XRD Spectra of Methylcellulose polymer electrolyte films.

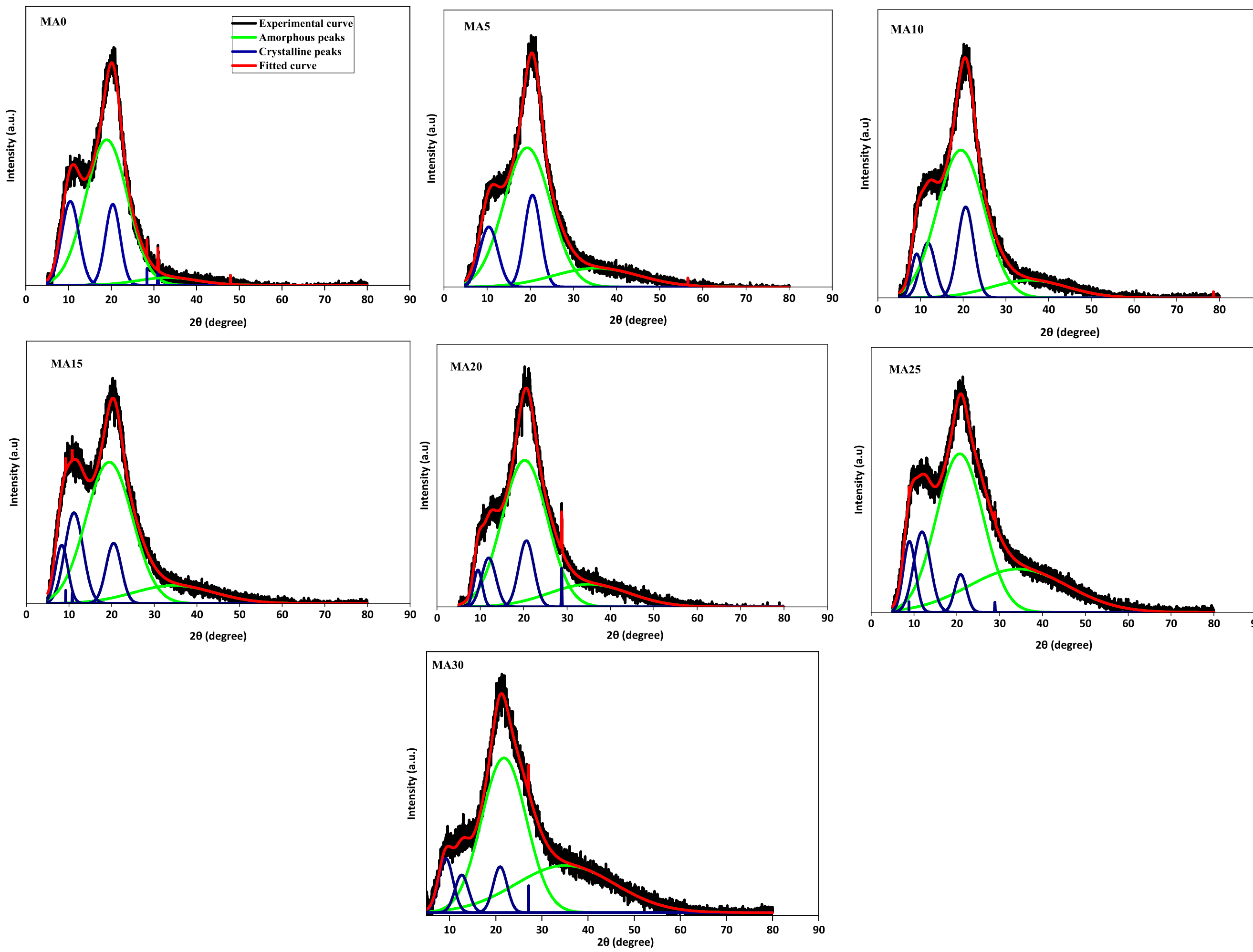

Figure 3. XRD deconvoluted graphs peaks for the polymer electrolytes.

Table II. Values of the degree of crystallinity of different polymer electrolytes.

Electrolyte	A_a	A_c	Degree of Crystallinity (χ_c) (in %)
MA0	9008.4	3614.9	28.8
MA5	11037.5	3504.7	24.1
MA10	12930.2	3672.9	22.1
MA15	10002.0	2674.3	21.1
MA20	9279.6	2264.2	19.6
MA25	14224.8	3241.1	18.6
MA30	9651.1	1608.6	14.3

group interacts with the NO_3^- anion. Considering these interactions, Fig. 5 illustrates a probable interaction pattern between the polymer and the salt.

SEM analysis.—Figure 6 depicts the Scanning Electron Microscopy images of the pristine polymer and the salt-incorporated electrolyte systems. Micrograph of MA0 is completely homogenous, indicating a smooth surface of the polymer film. With the inclusion of the salt, the surface becomes slightly rough with the formation of small shapes and white dots on the surface. This indicates the accumulation of charges on the surface of the polymer owing to the enhancement of ionic conductivity as well as transport parameters of the system.[10,13,22]

Impedance analysis.—A widely used technique to assess ionic conductivity in materials is impedance spectroscopy. Electrolyte samples were then characterized using impedance measurements. The resulting Cole-Cole plots (Fig. 7a) exhibit a high-frequency semicircle, which is indicative of bulk ionic transport within the material. Additionally, a low-frequency spike is observed, which is characteristic of the electrical double layer formed at the electrode-electrolyte interface.[23] Deviation of the Cole-Cole plot from the typical nature is due to the presence of inhomogeneities on the surface.[24] The impedance response was analyzed using an equivalent circuit model to extract the bulk resistance (as shown in Fig. 7b). The electrolyte systems ′ bulk conductivity (σ_{DC}) was then calculated using Eq. 2,[25] and the values are tabulated in Table IV.

$$\sigma_{DC} = \frac{t}{R_b A} \quad [2]$$

Here, t is the thickness of the film, and R_b and A refer to the bulk resistance and area of the film, respectively. Analysis of Table IV reveals a positive correlation between ionic conductivity and salt concentration in methylcellulose. The observed increase in mobile ion conductivity likely stems from a higher concentration of dissociated salts, as evidenced by the progressive amorphization of the electrolyte matrix revealed by XRD analysis. This suggests a breakdown of long-range order within the matrix at higher salt contents, potentially facilitating ion transport. This amorphization process enhances the flexibility of the polymer matrix, facilitating ion transport and, consequently, boosting ionic conductivity.[10] The ionic conductor, MA30, formulated with 30 wt% salt content,

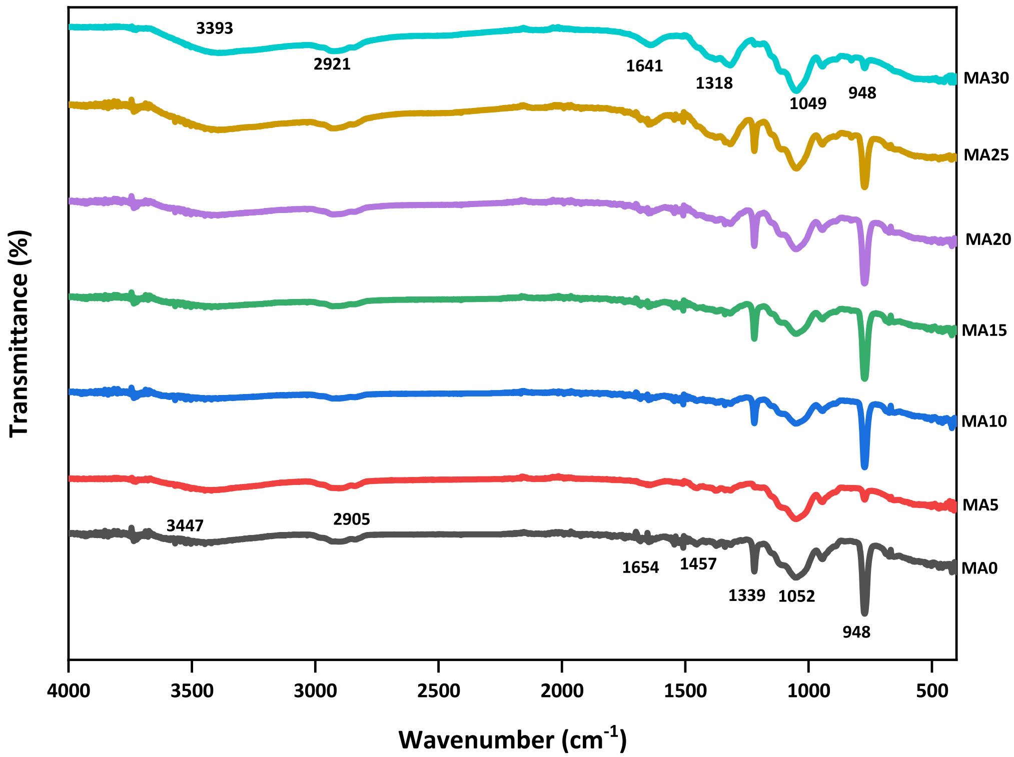

Figure 4. FTIR spectra of the prepared electrolytes.

Table III. Peak assignments of the vibrations of different functional groups of the electrolytes.

MA0	MA5	MA10	MA15	MA20	MA25	MA30	Peak Assignments
3447	3421	3503	3448	3421	3393	3393	O−H asymmetric stretching
2905	2904	2928	2928	2925	2917	2921	C−H asymmetric stretching
1649	1648	1648	1648	1648	1648	1641	O−H bending
1453	1454	—	—	—	—	—	The vibration of deformation in δ(C−H)
1339	1377	1339	1339	1339	1318	1318	C−H bending
1052	1051	1052	1051	1052	1048	1049	C−O−C bending
948	948	948	948	948	948	948	Glycosidic linkage

demonstrates a peak ionic conductivity of 3.78×10^{-5} S cm^{-1} at ambient temperature.

Dielectric Properties

The complex permittivity of the electrolyte systems can be represented as:[26]

$$\varepsilon^*(\omega) = \varepsilon' - j\varepsilon'' \qquad [3]$$

Material properties ε' and ε'' denote the dielectric constant and dielectric loss, respectively. The dielectric constant quantifies the extent of charge gathering at the electrode-electrolyte interface, considering both inherent and induced charges. In contrast, dielectric loss signifies the energy wasted as heat due to the movement of polar molecules within the material when subjected to an external electric field.[27] Both can be represented as:

$$\varepsilon'(\omega) = \frac{C_p t}{\varepsilon_0 A} \text{and } \varepsilon''(\omega) = \varepsilon' \tan\theta \qquad [4]$$

C_p, t, ε0, and A are the capacitance, thickness, permittivity of free space, and area of the electrolyte films. The dependence of the logarithmic value of dielectric constant (ε') and dielectric loss on the logarithm of angular frequency (log ω) is presented in Fig. 8. All materials exhibit a higher ε' at lower frequencies, followed by an exponential decrease at higher frequencies. This behavior can be attributed to the response of the material's dipoles to the applied electric field. At lower frequencies, both permanent and induced dipoles have sufficient time to orient themselves in the direction of the field, leading to a greater overall polarization and a higher ε'. However, as the frequency increases, the electric field reverses direction more rapidly. The dipoles are unable to respond as quickly, resulting in a decrease in the effective polarization and consequently, a lower ε'. This trend continues until a plateau is reached at very high frequencies, indicating that the dipoles are essentially immobile relative to the rapidly alternating field.[27,28] The data suggests a positive correlation between salt inclusion and the dielectric constant. This trend aligns with the increased concentration of dissociated ions at higher salt content. As the salt concentration rises, more ions become available, potentially enhancing the material's ability to store and release electrical energy.[29]

Dielectric loss spectra (Fig. 8) reveal a dependence on the applied electric field's angular frequency (logarithmic scale). Notably, all electrolyte systems exhibit a pronounced decrease in dielectric loss with increasing frequency, reaching a plateau at higher frequencies. This behavior can be attributed to the relaxation of polar entities within the polymer electrolyte matrix. At low frequencies, these dipoles have sufficient time to reorient themselves in response to the

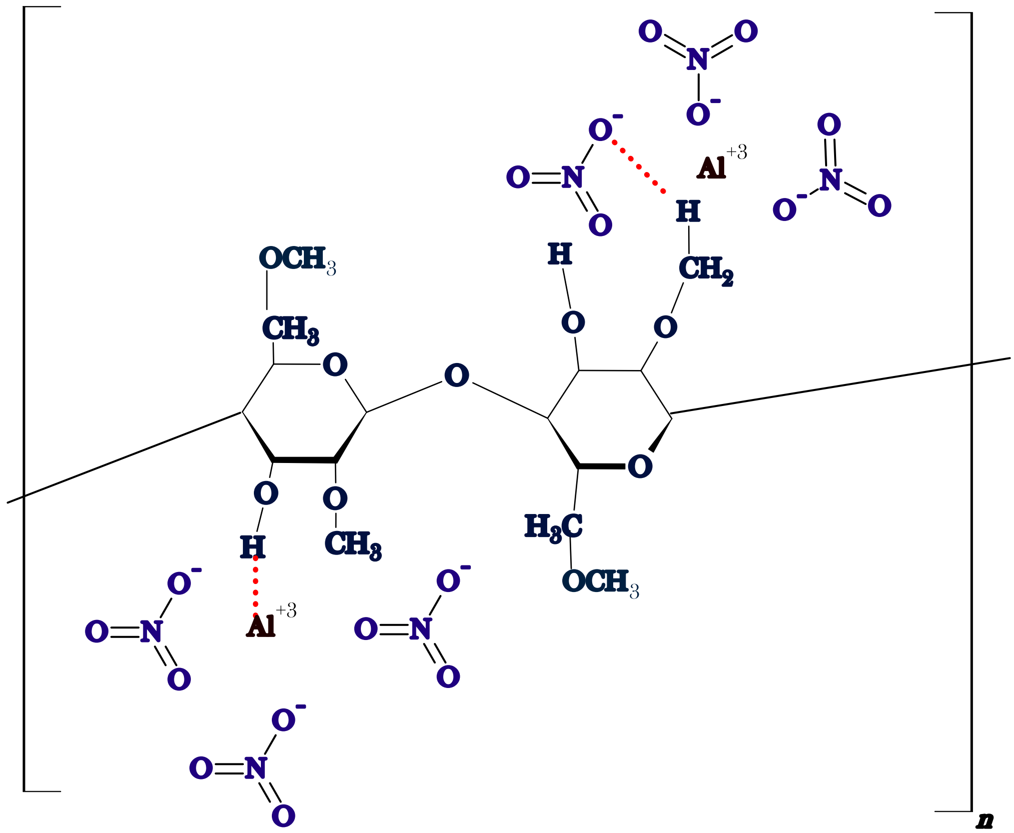

Figure 5. Schematic diagram of the probabilistic interaction between polymer and salt.

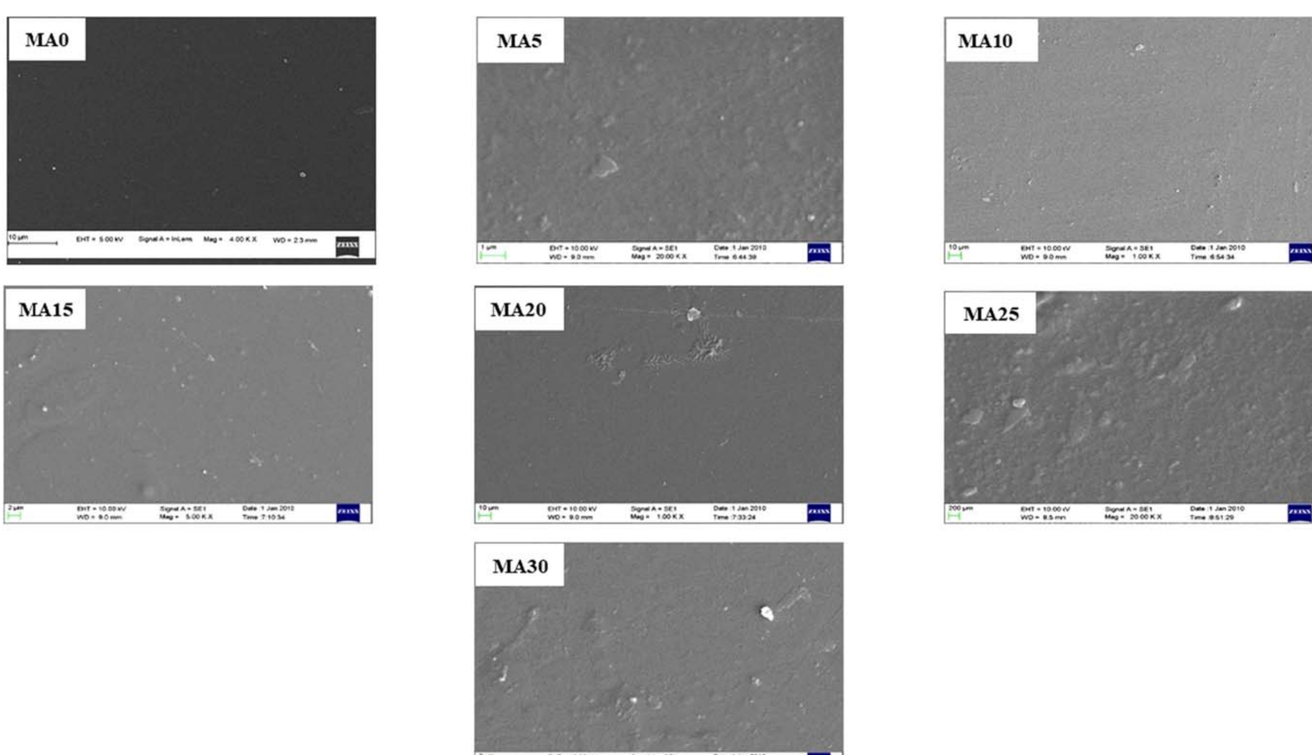

Figure 6. SEM micrographs of methyl cellulose and salt incorporated electrolyte systems.

alternating electric field, contributing significantly to dielectric loss (ε''). However, as frequency increases, the dipoles struggle to keep pace with the field's rapid fluctuations, leading to a decline in ε''. Furthermore, the data suggests a positive correlation between salt concentration and dielectric loss. This can be explained by the increased availability of free ions due to enhanced salt dissociation,

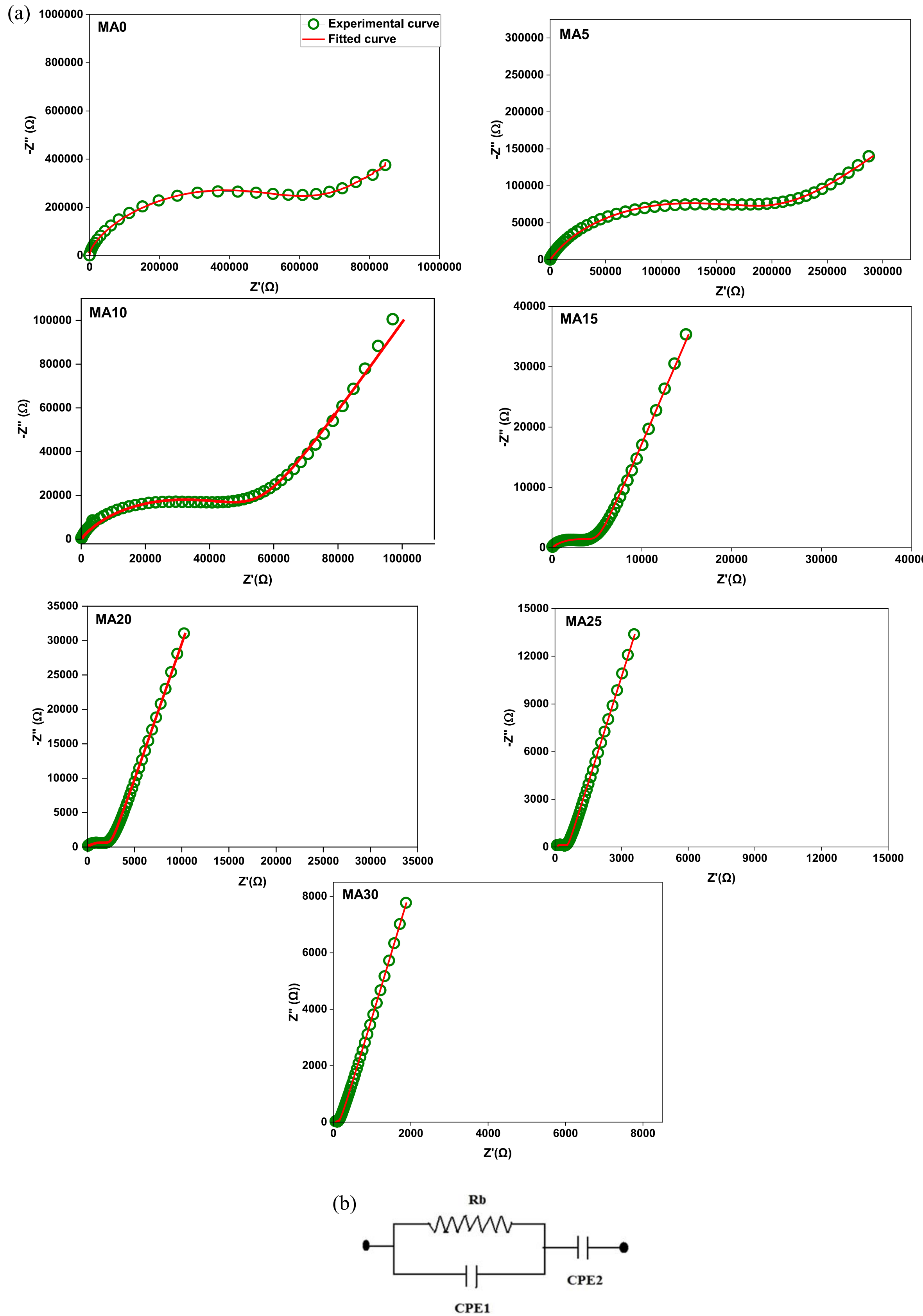

Figure 7. (a): The Cole-Cole plot of polymer electrolyte systems (b): Equivalent circuit used to fit the Cole-Cole plots.

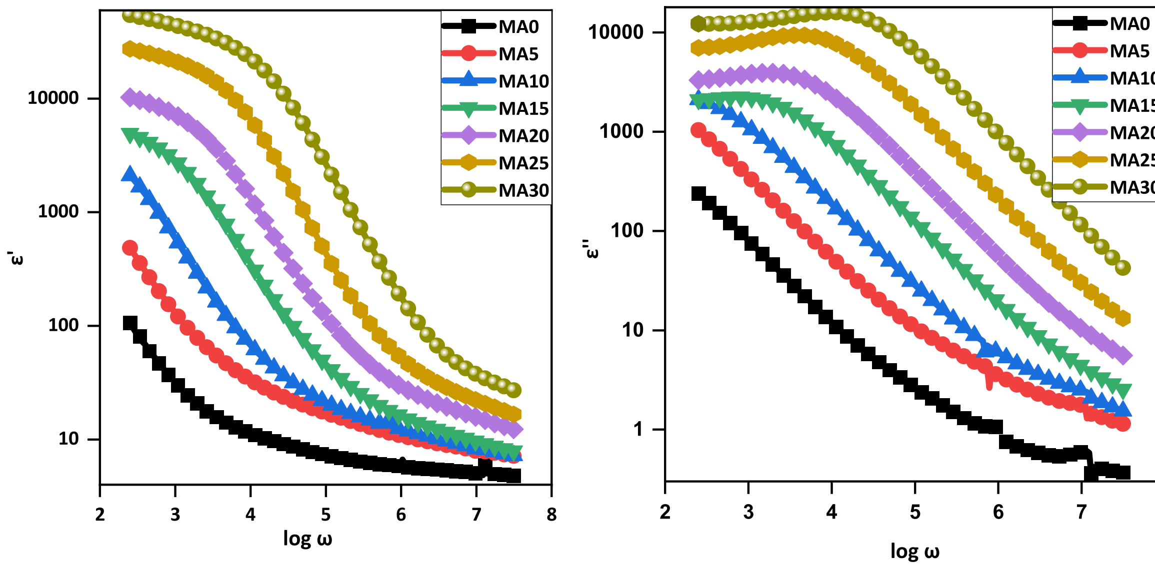

Figure 8. Variation of dielectric properties with log ω.

Table IV. Values of bulk resistance and ionic conductivity of different polymer electrolytes.

Electrolytes	Bulk resistance (Ω)	Thickness (μm)	Bulk conductivity (S/cm)
MA0	5.58×10^{5}	6.24×10^{-5}	9.89×10^{-9}
MA5	1.83×10^{5}	9.20×10^{-5}	4.44×10^{-8}
MA10	53767.00	1.03×10^{-4}	1.70×10^{-7}
MA15	5211.00	4.98×10^{-5}	8.46×10^{-7}
MA20	2578.00	8.42×10^{-5}	2.89×10^{-6}
MA25	602.64	9.22×10^{-5}	1.35×10^{-5}
MA30	242.78	1.04×10^{-4}	3.78×10^{-5}

facilitating ionic conduction and contributing to energy dissipation.[30,31]

Tangent Loss Analysis

Tangent loss is often associated with dielectric materials. In the case of polymer electrolytes. tan δ represents the ratio of energy lost to stored energy[32] As shown in Fig. 9, tan δ varies with the logarithm of angular frequency (log ω). The trend suggests an increase in tan δ with frequency, reaching a peak before another rise at higher frequencies. This peak, signifying maximum energy loss, shifts towards higher frequencies as the salt concentration increases. This shift can be attributed to enhanced ionic mobility within the polymer matrix, indicating faster ion hopping rates between available sites.[32] Equation 5 was employed to determine the relaxation times within the electrolyte systems. The data for these relaxation times are presented in Table V.

$$\tau_{\min} = \frac{1}{\omega_{\max}} \quad [5]$$

Relaxation time a system indicates the time required for the hopping of ions. The slower the relaxation time, the faster the ion movement.[33] Table V shows the decrease in the $\tau_{\max}$ with an increase in the salt concentration. MA30, an electrolyte system with 30 wt% of aluminium salt, shows a low relaxation time and hence exhibits high ionic conductivity due to the faster conduction of ions.

AC Conductivity Analysis

The presented figure (Fig. 10) depicts the AC conductivity spectra obtained for the synthesized polymer electrolytes. These spectra characteristically display three distinct regimes. At low frequencies, a dispersion region is observed. This can be attributed to the build-up of ions at the interface between the electrodes and the electrolyte, a phenomenon well-understood as the electrical double layer effect. In the mid-frequency range, a plateau emerges in the spectra. This plateau signifies that the dominant mechanism for conductivity becomes the long-distance migration of ions within the material. Finally, at high frequencies, another dispersion region is observed. This can be explained by the short-range movements of ions confined within the bulk of the electrolyte.[34] All the systems show three distinct regions. A short high-frequency dispersion region for higher salt concentrations indicates the hopping in the system.[34]

Analysis of the prepared polymer electrolytes using alternating current (AC) conductivity spectroscopy reveals adherence to Jonscher's universal power law, as described by the equation:[35]

$$\sigma_{ac}(\omega) = \sigma_{dc} + A\omega^{n} \quad [6]$$

This notation defines the key parameters: ω represents the angular frequency, a critical factor in oscillatory phenomena. The key parameter "A" captures how temperature influences the system, and n, the exponent, defines the power-law relationship, with values constrained between zero and one,[36,37] $\sigma_{ac,}$ and σ_{dc} are the AC conductivity and DC conductivity of the system, respectively. AC conductivity spectra can be a valuable tool for extracting DC conductivity information in electrolyte systems. This can be achieved through two main approaches: employing Jonscher's power law for spectral fitting or extrapolating the mid-frequency plateau region to the vertical axis (conductivity). In this study, we opted for the latter method to determine DC conductivity. The obtained values, presented in Table V, exhibit good agreement with the DC ionic conductivity measurements performed using impedance spectroscopy.

Ion Transport Properties Analysis

Ion transport properties encompass the characteristics and behaviors associated with ion movement within a material. These properties are particularly relevant for electrolytes, as they dictate the material's ability to conduct electricity via ionic flow.

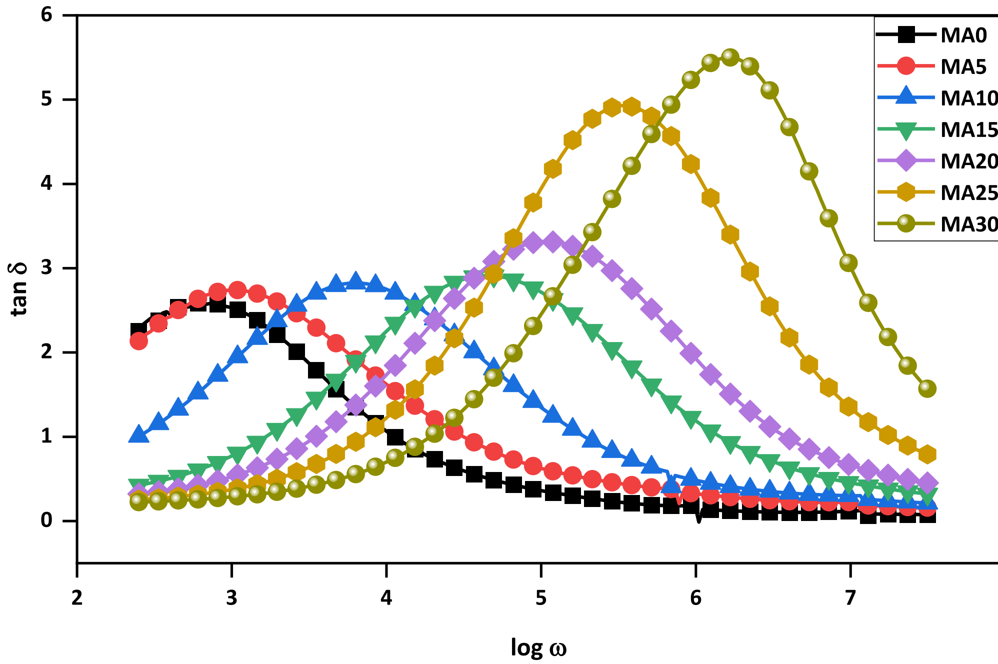

Figure 9. Variation of loss tangent with log ω.

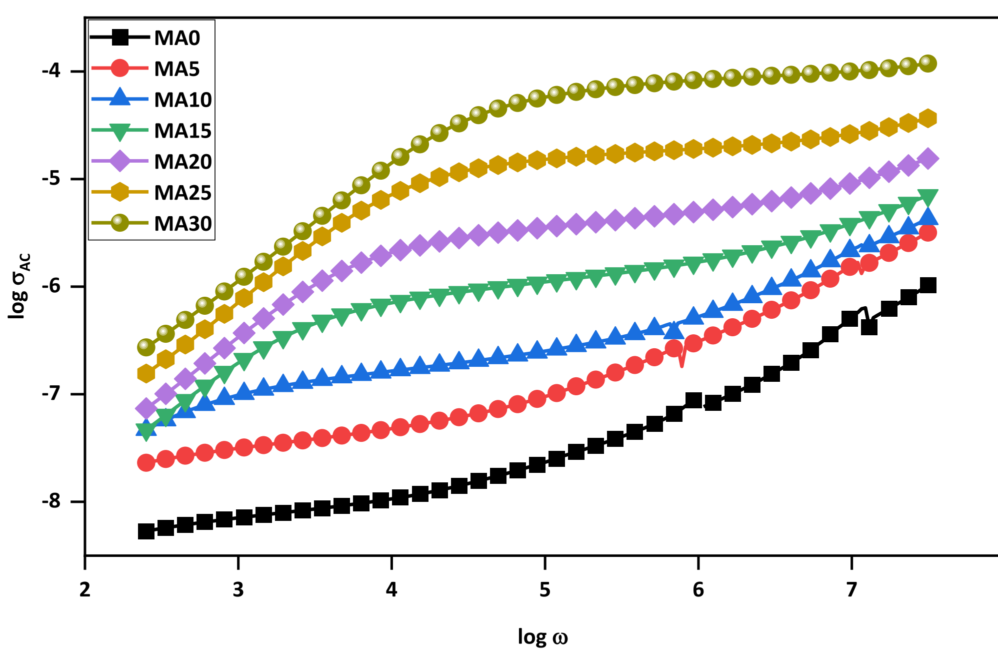

Figure 10. AC conductivity spectra of the prepared electrolytes.

Understanding ion transport is critical across various scientific and technological fields, including electrochemistry, material science, and energy storage applications. Electrochemical impedance spectroscopy (EIS) serves as a valuable tool for extracting transport parameters such as diffusion coefficient (D), mobility (μ), and number density (n). In this study, EIS data was analyzed by fitting the Cole-Cole plot for the polymer electrolyte using dedicated EIS software.[3] Subsequently, n, μ, and D values were determined using the following formulas.

$$n = \frac{\sigma_{dc} kT\tau_2}{(ek_2\varepsilon_r\varepsilon_0 A)^2} \quad [7]$$

$$\mu = \frac{e(k_2\varepsilon_r\varepsilon_0 A)^2}{kT\tau_2} \quad [8]$$

$$D = \frac{(k_2\varepsilon_r\varepsilon_0 A)^2}{\tau_2} \quad [9]$$

Here, k_2 is the inverse of the capacitance equivalent to the effective double-layer capacitance, εr is the dielectric constant of the system at a frequency of 1 MHz, ε0 is the permittivity of free space (8.85×10^{-12} F m^{-1}), A is the area of the electrolyte, k is

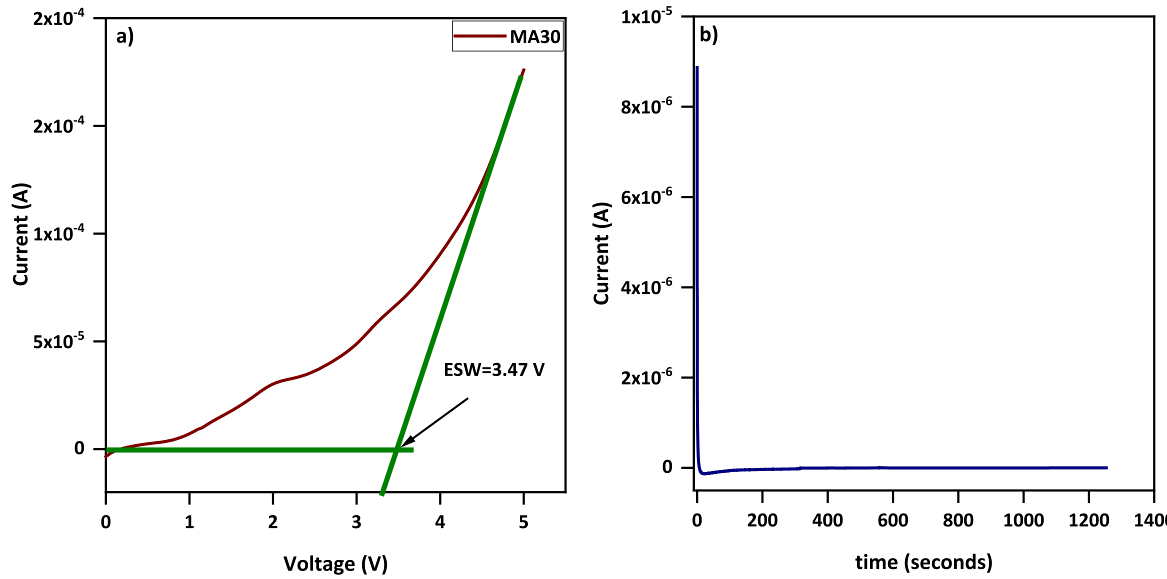

Figure 11. (a) I-V characteristics of MA30, (b) DC polarization curve of MA30.

Table V. Values of relaxation time and DC conductivity values obtained from AC conductivity spectra.

Electrolyte	Relaxation time (seconds)	DC conductivity at $\omega = 0$
MA0	1.650×10^{-3}	5.13×10^{-9}
MA5	9.735×10^{-4}	2.29×10^{-8}
MA10	1.579×10^{-4}	1.90×10^{-7}
MA15	2.277×10^{-5}	8.31×10^{-7}
MA20	9.442×10^{-6}	3.02×10^{-6}
MA25	2.920×10^{-6}	1.38×10^{-5}
MA30	5.988×10^{-6}	4.89×10^{-5}

Table VI. Values of n, μ, and D of various polymer electrolytes.

Electrolyte	Diffusion coefficient D (cm^2 s^{-1})	Mobility μ (cm^2 V^{-1} s^{-1})	Number density n (cm^{-3})
MA5	3.44×10^{-7}	1.32×10^{-5}	2.11×10^{16}
MA10	5.63×10^{-7}	2.15×10^{-5}	4.92×10^{16}
MA15	8.46×10^{-7}	3.25×10^{-5}	1.63×10^{17}
MA20	3.41×10^{-8}	1.30×10^{-6}	1.39×10^{19}
MA25	3.89×10^{-9}	1.47×10^{-7}	5.75×10^{20}
MA30	2.22×10^{-9}	8.50×10^{-8}	2.78×10^{21}

Boltzmann's constant (1.38×10^{-23} J K^{-1}), $\tau_2 = 1/\omega_2$, where ω_2 is the angular frequency corresponding to the minimum value of the imaginary part of impedance, and e is the electronic charge (1.61×10^{-19} C).[3] The values of n, μ, and D, along with the ionic conductivity of different polymer electrolytes, were given in Table VI.

The table above makes it evident that the number density of polymer electrolytes rise with an increase in salt concentration. Aluminium nitrate dissolution generates free aluminium and nitrate ions. These ions likely complex with the polymer's hydroxyl (-OH) and methyl (-CH) groups, leading to an increase in the number density of species within the material. Introducing the salt promotes further dissociation, resulting in a higher concentration of charge carriers.[19] The interplay between solvent molecules, dissolved ions, and the polymer matrix in electrolytes significantly affects ion transport. Studies have shown that initially, increasing salt concentration leads to enhanced diffusion coefficients and ionic mobility within the electrolyte system. However, this trend appears to reverse at higher salt concentrations. This phenomenon might be attributed to a stiffening effect on the polymer chains, hindering ion movement.[38] The relationship between bulk ionic conductivity (σ) and both the number density (n) and mobility (μ) of ions is well established ($\sigma = n\mu e$). This study reveals that a decrease in diffusion coefficient and mobility can be counterbalanced by a significant rise in number density. This phenomenon ultimately leads to an enhancement of the ionic conductivity within the material.

Electrochemical stability window (ESW) and ion transference number measurement.—Figure 11 presents the current-voltage (I-V) and current-time (I-t) characteristics of the MA30 electrolyte system, exhibiting the highest conductivity among the investigated samples. The I-V data were collected using a two-electrode configuration with stainless steel electrodes sandwiching the MA30 electrolyte film.

The electrochemical stability window (ESW) is a critical parameter for electrolyte selection in energy storage applications. It was determined by extrapolating the linear region of the I-V curve to the voltage axis, resulting in an ESW of 3.47 V for the MA30 electrolyte. This wide ESW signifies the excellent stability of the electrolyte and its potential suitability for various energy storage devices.

The study employed an electrochemical characterization technique, specifically Wagner's DC polarization, to determine the transport number within the MA30 electrolyte. This technique involved placing the electrolyte between blocking electrodes (stainless steel in this case) and monitoring the current response over time.[24,25] Figure 11b illustrates the temporal dependence of electrical current. The following equation was employed to calculate the transference number of ions[10,39] with i_i and i_f representing the initial and final current through the cell:

$$t_{ion} = \frac{i_t - i_f}{i_i} \quad [10]$$

Analysis of the ion transference number (0.999) suggests near-exclusive ionic conduction within the material. This implies a dominance of ionic species as the charge carriers.

Fabrication of an aluminium ion conducting primary battery.—The increasing demand for portable electronics, electric vehicles, and renewable energy sources has intensified the need for improved primary battery materials. While these batteries are

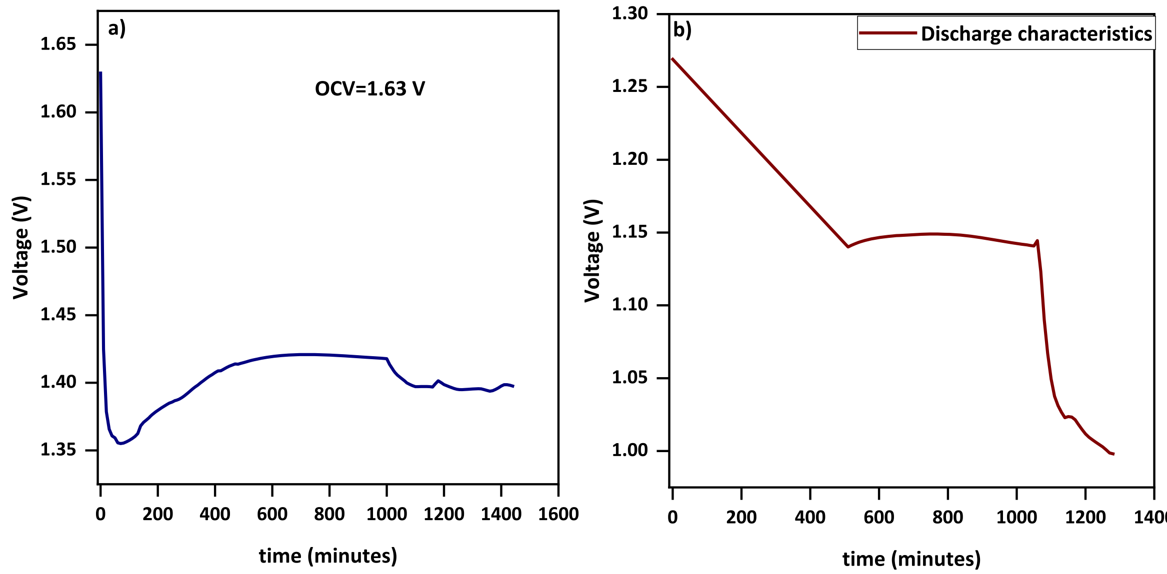

Figure 12. (a) Open circuit potential of the prepared battery, (b) Discharge characteristics of the battery.

Table VII. Cell parameters across 100 kΩ resistor.

Cell parameters	Measured values of discharge across 100 kΩ
Cell area (cm^2)	1.54
Cell weight (g)	1.96
Effective cell diameter (cm)	1.2
Cell height (mm)	6.61
Open circuit voltage (V)	1.63
Current drawn (μA)	50
Discharge time plateau region (mins)	540
Energy density (mWh/kg)	12.18
Power density (mW/kg)	41.6

valued for their reliability and simple design, their current limitations in energy density and the environmental impact of traditional electrolyte materials are significant roadblocks to wider utilization. This study examines the performance of a primary battery employing a newly developed electrolyte (MA30) as its anode. Vanadium pentoxide (V_2O_5), graphite powder, and MA30 electrolyte composite were prepared for the counter electrode using a 3:1:1 ratio via a grinding process. The mixture was compressed into a pellet using a pressure of 5 tons. Aluminum metal served as the current collector (anode). The MA30 electrolyte was positioned between the cathode and anode within a battery enclosure.

The open-circuit voltage (OCV) and discharge behavior of the battery were evaluated and are presented in Fig. 12. The fabricated battery exhibited an initial OCV of 1.63 V. This voltage observed a decrease over time, eventually reaching a near-constant value of 1.4 V. The observed voltage reduction might be attributed to reactions occurring at the interface between the electrodes and the electrolyte, as represented by the following equation:

At the anode-

$$Al + 3(OH^-) \rightarrow Al(OH)_3 + 3e$$

At the cathode-

$$3V_2O_5 + 3H_2O + 2e \rightarrow 3V_2O_4 + 6OH^-$$

Thus, the overall reaction would be: -

$$Al + 3V_2O_5 + 3H_2O \rightarrow Al(OH)_3 + 3V_2O_4$$

The polymer matrix acts as a source of OH^- ions. Figure 11b depicts the discharge behavior of the fabricated battery when connected across a 100 kΩ resistor. The observed voltage reduction to 1.27 V upon connection is likely due to the polarization effect.[40] This effect diminishes over time, reaching a steady state value of around 1.14 V after 7 h. A plateau region in the discharge curve is observed for 9 h, followed by a further decrease in voltage. Table VII summarizes the key characteristics of the prepared cell.

Conclusions

This study investigated a new methylcellulose-based solid polymer electrolyte system doped with aluminium nitrate salt. The electrolyte films were fabricated using a solution-casting technique. Various characterization methods were employed to understand the films' properties. X-ray diffraction (XRD) analysis revealed that the MA30 electrolyte film exhibited the lowest crystallinity, approximately 14.29%, indicating a highly amorphous structure. Fourier-transform infrared spectroscopy (FTIR) analysis confirmed the successful complexation between the polymer and the salt, as evidenced by the distinct vibrational peaks observed in the spectra. Electrochemical impedance spectroscopy (EIS) measurements identified MA30 as the most conductive film, with a conductivity of around 3.78×10^{-5} S cm^{-1}. This observation aligns with the expected trend of increasing conductivity with a higher salt concentration within the polymer matrix. The highest conducting electrolyte (MA30) demonstrated a promising electrochemical stability window (ESW) of 3.47 V and a high ionic transference number of 0.999, indicating efficient ion transport within the film. Finally, a primary battery fabricated using the MA30 film displayed good discharge characteristics, suggesting its potential application in energy storage devices.

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Author Contributions

Saranya Vijayan: Methodology, Formal analysis, Investigation, Writing - Original Draft. Ismayil: Conceptualization, Validation, Resources, Supervision, Writing - Review & Editing. Jayalakshmi Koliyoor: Formal analysis, Data Curation, Writing - Review & Editing.

Data and Code Availability

The data supporting this study's findings are available on request from the corresponding author.

Ethical Approval

This study was carried out following all ethical standards; neither human participants nor animals were used during the study.

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